

Пробная олимпиада ISTCiM – 3

Time allowed: 4 hours

14 сентября 2026 года

A. Algebra and combinatorics. Suppose that all complex roots of a polynomial with integer coefficients and leading coefficient 1 have absolute value 1. Prove that all of them are roots of unity.

C. Calculus and mathematical analysis. Let $a_0 = 5/2$ and $a_k = a_{k-1}^2 - 2$ for $k \geq 1$. Compute

$$\prod_{k=0}^{\infty} \left(1 - \frac{1}{a_k}\right)$$

in closed form.

E. Equations and inequalities. Let m and n be positive integers. Prove that

$$\frac{(m+n)!}{(m+n)^{m+n}} < \frac{m!}{m^m} \cdot \frac{n!}{n^n}.$$

G. Geometry and linear algebra. What is the smallest possible rank of an $n \times n$ real matrix whose diagonal entries are zero and whose off-diagonal entries are positive?

P. Set theory and probability. Let $x_0 = 1$ and let $\delta \in (0,1)$ be a constant. For $n = 0, 1, 2, \dots$ a point x_{n+1} is chosen uniformly at random from the interval $[0, x_n]$. Let Z be the least n for which $x_n < \delta$. Find the expected value of Z as a function of δ .